3.13.4 Operational amplifier in:

3.13.4.1 inverting amplifier configuration (A-level only)

Content
Derivation of $\frac{V_{\mathrm{out}}}{V_{\mathrm{in}}} = -\frac{R_{\mathrm{f}}}{R_{\mathrm{in}}}$, calculations. Meaning of virtual earth, virtual-earth analysis.

$$\frac{V_{out}}{V_{in}} = 1 + \frac{R_f}{R_1}$$

Derivation is not required.

3.13.4.3 summing amplifier configuration (A-level only)

Content
$V_{out} = -R_f\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3} + \ldots\right)$ Difference amplifier configuration. Derivation is not required. $V_{out} = \left(V_+ - V_-\right)\frac{R_f}{R_1}$ Derivation is not required.

3.13.4.4 Real operational amplifiers (A-level only)

Content
Limitations of real operational amplifiers. Frequency response curve. $gain \times bandwidth = constant$ for a given device.